serial bulbs. In serial circuit, the path of the current flow is only one. So if any fault (brake in wire connection) occurs in any one place of the circuit, the total circuit (serial set) is inactive.

2.6.2 Parallel Circuit

When resistors are connected across one another so that the same voltage (supply voltage) is applied between the end point of each resistor, then they are said to be in parallel connection.

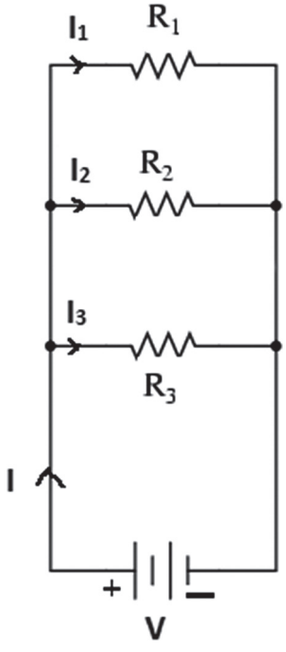


Fig 2.6 *Parallel Circuit*

In this circuit, the voltage across each resistor is same as supply voltage but the current in each resistor is different. In this circuit, the sum of the current I_1, I_2 and I_3 is equal to supply current I, i.e $I = I_1 + I_2 + I_3$

According to ohm's law

$$I = \frac{V}{R}$$

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}$$

But $I = I_1 + I_2 + I_3$

$$I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]$$

$$\frac{I}{V} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{I}{V} = \frac{1}{R}$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{R_2R_3 + R_1R_3 + R_1R_2}{R_1R_2R_3}$$

$$R = \frac{R_1R_2R_3}{R_2R_3 + R_1R_3 + R_1R_2}$$

Important rules of a parallel circuit

i) In parallel circuit, the current flows through two or more parallel paths at a junction.

ii) Current varies in different resistances i.e The sum of the current is equal to supply current.

$$I = I_1 + I_2 + I_3 + \ldots$$

iii) The voltage is same in all resistors as supply voltage.

$$V = V_1 = V_2 = V_3 = \ldots$$

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